

Real-Time Longitudinal and Lateral State Estimation of Preceding Vehicle Based on Moving Horizon Estimation

Hanghang Liu¹, Ping Wang¹, Jiamei Lin, Haitao Ding¹, Hong Chen², and Fang Xu¹

Abstract—In the advanced driver assistance system (ADAS) and autonomous driving systems, accurate information on preceding vehicle motion states is vital for the path planning and control. To develop those intelligent driving systems, a modular integrated estimation algorithm for the preceding vehicle longitudinal and lateral states is proposed in this paper; the coupled nonlinear characteristics of vehicle dynamics are applied to improve state estimation accuracy. First, considering driver aggressiveness, a linear moving horizon estimator for the vehicle longitudinal speed is designed based on the car following model. Then, the estimated vehicle longitudinal speed is delivered to the lateral estimator module in real-time. For the arbitrary driving routes, the Serret-Frenet equations and nonlinear lateral dynamics are combined to describe the lateral motion of preceding vehicle. Furthermore, a nonlinear moving horizon estimator is constructed to estimate vehicle lateral states accurately. To reduce the computational burden, the multiple shooting (MS) method is introduced, then the objective function is divided into several segments to improve the strong nonlinearity produced by multiple iterations. The proposed estimation algorithm can effectively handle the initial biases, the additive biases and model error, which are common in the acquired signals, and provide an accurate estimate of preceding vehicle states.

Index Terms—Advanced driver assistance system (ADAS), state estimation, moving horizon estimation (MHE), multiple shooting (MS).

I. INTRODUCTION

IN RECENT years, with the continuous increase in vehicles and notable development of information and intelligent technology, the advanced driver assistance system (ADAS) and

autonomous driving system (ADS) have attracted wide interest [1]–[3]. In general, the ADAS and ADS can be summarized in three parts: perception, planning and control. The perception part collects and processes information from the ego vehicle and its surrounding environment. The processed information provides the necessary support to the planning and control parts. The perception part is the pioneering component in the whole system.

Regarding the perception part, there are many studies on the vehicle state estimation [4], [5]. The longitudinal motion states (usually are speed and acceleration) are widely used in many ADAS systems, it has a great significance to derive these information. For example, in adaptive cruise control and autonomous emergency braking systems, it is necessary to derive the preceding vehicle speed and the distance between the host and preceding vehicles [6]–[8]. In earlier studies, the Kalman filter (KF) was employed in vehicle position and speed estimation for the purpose of target tracking [9], [10]. Recently, with the development of sensors and relative technologies, the inter-vehicle distance can also be obtained based on the vision systems [11]. To accurately depict the relationship between the vehicle and its surroundings and improve the estimation accuracy, it is necessary to model the vehicle motion. Helbing and Molnar proposed a social force model (SFM) to depict the interaction relationships between pedestrians [12]. Then, the SFM is modified to model the vehicle dynamics and a vehicle tracking method was proposed based on the modified SFM in [13], which achieved a higher accuracy. However the SFM is only an empirical approximate model, and is not accurate enough to precisely describe the vehicle motion. Substantial studies have also focus on the car following model, which can precisely depict the following vehicle motion for designing controllers and estimators [14]. In [15], a new car following model (CFM) based on the Helly's model was proposed to describes the dependent motion of vehicles in the single-lane case. Then, combined with Kalman filter (KF), a multivehicle motion estimation algorithm was proposed that attained a satisfactory performance for vehicle speed estimation.

In fact, it is not enough to observe only the longitudinal motion. For example, when a host vehicle is about to overtake a preceding vehicle, the preceding vehicle lateral motion can be used to decide whether the overtaking process should be terminated [16]. In other scenarios, such as trajectory planning

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and car-following control, the lateral motion states are also critical [17], [18]. Although, the motion states may be measurable for the lidar and high accuracy radar. The high costs caused by these high performance sensors are still unacceptable for the auto makers. For the sensors equipped in production vehicles, the preceding vehicle lateral motion states are very difficult to directly measure. With the development of vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I) communications and vehicle recognition technology [18]–[21], it is possible to transmit some intrinsic parameters including mass, moment of inertia, cornering stiffness and wheelbase between vehicles. Besides, in vehicle platoon and following control, the dynamic vehicle information such as longitudinal and lateral motion states and location can also be transmitted by V2V and V2I. S. Wei *et al.* [18] presented an integrated longitudinal and lateral vehicle following control system where the V2V was used to transmit the preceding vehicle states and parameters. However, the V2V and V2I are easily influenced by the communication delays and disturbances and thus not suitable to transmit the complete longitudinal and lateral motion states. Besides, the geometry of the roads parameters should also be incorporated in the model. In the study presented in [22], a stochastic modeling method was proposed to create realistic artificial road geometry, where the number of segments, length of road segments, left and right bend deflection degrees and the height of up and down hills are generated by the Poisson, exponential, Gaussian and Rayleigh distributions, respectively. In [23], the steady-state responses of a bicycle-roll vehicle model was analyzed to give the instantaneous center of rotation (ICR) under the constant velocity and steering angle. Then, the desired velocity and steering angle can be derived based the curvature center of road. Y. Wang *et al.* [16] proposed a combined vehicle model based on the vehicle dynamic model and road geometry constraints. Then a KF-based lateral state estimator was also proposed in this study. However, the proposed filter [16] was designed under the assumption that the preceding vehicle longitudinal speed and accurate parameters can be derived by vehicle recognition technology in advance. This assumption is difficult to satisfy under the current technology, which limits the application in vehicles. Besides, the vehicle is easy to enter the nonlinear regions under the extreme conditions [24]. In this case, the nonlinearity of vehicle must be considered so that the estimation accuracy can be improved.

The aforementioned estimation algorithms are mostly designed based on KF and its extension. Indeed, the KF has been applied in many fields for its simplicity and effectiveness. However, in certain cases, such as when a large initial error and strong nonlinearity occur, KF-based algorithms can not attain a satisfactory performance. When the system is highly nonlinear, the moving horizon estimation (MHE) method could be a competitive choice [25]. In addition, MHE also tends to be more robust and adaptive to model and parameter uncertainties, because multiple groups of measurements are used in the estimation. For the above reasons, MHE has already been applied in vehicle parameters and state estimation [26]–[28]. In addition, the superiority of MHE has been verified [28], in which the MHE based estimator attains a higher accuracy, higher

convergence speed with a large initial error and higher robustness to disturbance. However, MHE-based estimators require a long computational time, which is the largest barrier for MHE applications. To solve this problem, on the one hand, with the development of hardware, the computational ability is rapidly improving, while on the other hand, the optimization algorithm can also be improved, such as the direct MS method.

Compared with the existent studies, the main contributions of the proposed estimation algorithm are as follows:

(1) Unlike the conventional studies assuming the longitudinal speed of preceding vehicle is known, a linear MHE based estimator for the longitudinal speed derivation is developed. The Helly's SFM is incorporated in the longitudinal model, so that both the driver and model uncertainties can be considered in the proposed estimator. Then, the estimation results are provided to update the longitudinal speed in the lateral estimator.

(2) To satisfy the demand of information perception for intelligent vehicle under all driving conditions, the nonlinearity of vehicle must be taken into account. In the conventional preceding vehicle lateral state estimation studies, linear vehicle models are adopted to depict the vehicle dynamics. To achieve a higher estimation accuracy, the nonlinearity of vehicle motion are considered, then combined with the MHE algorithm, a lateral states estimator is constructed for the high estimation accuracy.

(3) One of the biggest obstacles prevent MHE from been applied in vehicular environment is the heavy computational burden caused by nonlinear optimization. To solve this problem, the multiple shooting method is adopted for the optimization problem formulation. With the MS method, the objective function is redesigned, then the computational burden can be reduced obviously and makes it possible to implement MHE in a vehicular environment.

The rest of this paper is organized as follows. The related models used to design the estimators are introduced in Section II. The design procedure of the estimators with the MS method as well as the guidelines for estimator parameter selection are introduced in Section III. Hardware in loop (HIL) experiments and results are provided in Section IV, and conclusions are summarized in Section V.

II. RELATED MODELS

A. Vehicle Longitudinal Motion Model

Here the CFM of Helly [33] is adapted to describe the vehicle longitudinal motion. Under the car-following scenario, the host vehicle acceleration a_h is given as:

$$a_h(k) = C_1(v_p(k) - v_h(k)) + C_2d(k) + C_3v_h(k) + C_4, \quad (1)$$

where $v_h(k)$, $v_p(k)$ and $d(k)$ are the host vehicle longitudinal speed, preceding vehicle longitudinal speed and relative distance between the two vehicles, respectively, at time instant k . The coefficients C_1 , C_2 and C_3 can be determined with real experiments [33]. To reflect the different driving habits of different drivers, a coefficient C_4 called driver's aggressiveness, which determines the distance between the host and preceding vehicles when they reach a steady-state equilibrium (the acceleration of

following vehicle satisfies $a_h = 0$), is introduced to calculate the acceleration of host vehicle.

We assume that the host vehicle drives in the current lane and follows the preceding vehicle, and the host vehicle will be influenced only by the preceding vehicle. At this time, we define the system state as:

$$x(k) = [v_p(k), d(k), v_h(k)]^T. \quad (2)$$

Then we define the system sample time of the longitudinal module as δ_k . The system can be derived as:

$$x(k+1) = F(\delta_k)x(k) + \Gamma(\delta_k)[a(k) + \omega(k)], \quad (3)$$

where $a(k) = [0, a_h(k)]^T$ is the additional acceleration of the driver, and considering the host vehicle acceleration (1), we obtain:

$$a(k) = \Gamma_a x(k) + C, \quad (4)$$

where,

$$\Gamma_a = \begin{pmatrix} 0 & 0 & 0 \\ C_1 & C_2 & C_3 - C_1 \end{pmatrix}, C = \begin{pmatrix} 0 \\ C_4 \end{pmatrix},$$

The state transition matrix $F(\delta_k)$ and the acceleration gain matrix $\Gamma(\delta_k)$ are defined as:

$$F(\delta_k) = \begin{pmatrix} 1 & 0 & 0 \\ \delta_k & 1 & -\delta_k \\ 0 & 0 & 1 \end{pmatrix}, \quad (5)$$

$$\Gamma(\delta_k) = \begin{pmatrix} \delta_k & 0 \\ \frac{1}{2}\delta_k^2 & -\frac{1}{2}\delta_k^2 \\ 0 & \delta_k \end{pmatrix}. \quad (6)$$

It should be noted that, in Eq. (3), a two-dimensional vector process noise $\omega(k)$ is the noisy acceleration component to model the disturbances of driving behaviours in the system. Then, to facility the equation construction process, a zero additional acceleration is added in $a(k)$, so that the additional acceleration can be a two-dimensional vector to simplify the equation construction process.

By substituting Eq. (4) into Eq. (3), we attain:

$$x(k+1) = A(\delta_k)x(k) + \Gamma(\delta_k)[C + \omega(k)], \quad (7)$$

where,

$$A(\delta_k) = F(\delta_k) + \Gamma(\delta_k)\Gamma_a.$$

It should be noted that, different drivers may have different driving styles, which are subject to different aggressiveness parameters, and the aggressiveness parameters of drivers may not be known in advance. As such, the driver aggressiveness parameter should also be estimated. Therefore we expand the state vector as $\bar{x}(k) = [x^T(k), C^T]^T$, and considering Eq. (7), a new state space equation can be derived as:

$$\bar{x}(k+1) = \bar{F}(\delta_k)\bar{x}(k) + \bar{\Gamma}(\delta_k)\omega(k), \quad (8)$$

where,

$$\bar{F}(\delta_k) = \begin{pmatrix} A(\delta_k) & \Gamma(\delta_k) \\ 0 & I_{2 \times 2} \end{pmatrix}, \quad (9)$$

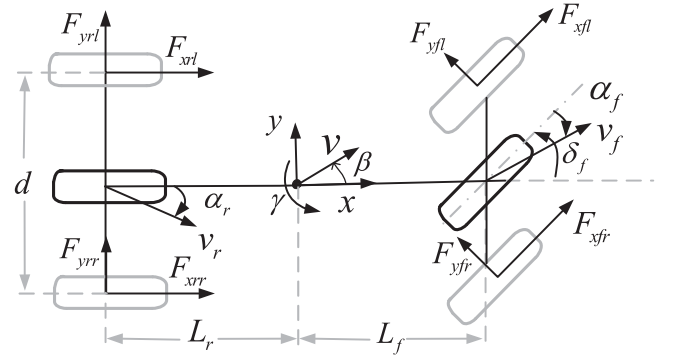


Fig. 1. The 2DOF bicycle vehicle model.

$$\bar{\Gamma}(\delta_k) = \begin{pmatrix} \Gamma(\delta_k) \\ 0 \end{pmatrix}. \quad (10)$$

The measurements of this system are the distance between the host and preceding vehicles d and the host vehicle velocity v_h . The relative distance can be derived based on radars and vision systems [1], [11], [18], and the host vehicle speed can be measured by sensors. However, sensor measurements are commonly polluted by noise, and measurement uncertainties must be considered. Therefore, we can derive the measurement equation as:

$$y(k) = D\bar{x}(k) + \eta(k), \quad (11)$$

$$D = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}. \quad (12)$$

Where D is the measurement matrix and $\eta(k)$ is the white Gaussian noise sequence which represents the measurement noise.

B. Vehicle Lateral Motion Model

As the main purpose of this paper is to estimate the lateral speed and yaw rate simultaneously, the lateral speed is highly coupled with the vehicle dynamics. To consider the vehicle dynamic characteristics, a 2-DOF yaw plane vehicle model is considered here. At the front and rear axles, the left and right wheels (the grey wheels in Fig. 1) are lumped into a single wheel (the black wheels in Fig. 1). The driver can only steer the front wheels, and the steering angles of the two front wheels are equal. Hence, the vehicle model is simplified as a bicycle model.

On the basis of the vehicle dynamics theory, the simplified vehicle dynamics are described as:

$$\dot{v}_y = \frac{F_{yf} + F_{yr}}{m} - \gamma v_x, \quad (13a)$$

$$\dot{\gamma} = \frac{L_f F_{yf} - L_r F_{yr}}{I_z}. \quad (13b)$$

Where the system states v_y and γ represent the lateral speed and yaw rate, the symbol F_{yf} and F_{yr} denote lateral tire forces of front and rear tires. The lateral tire forces are described by the single friction coefficient Fiala tire model [30]. This model uses the slip angle as an intermediate variable. When the sideslip angle α is very small, $\tan \alpha \approx \alpha$. Then, the tire model

TABLE I
PRECEDING VEHICLE VARIABLES AND PARAMETERS

Symbol	Description	Value	[Units]
v_x	Longitudinal speed	-	[m/s]
v_y	Lateral speed	-	[m/s]
γ	Yaw rate	-	[rad/s]
δ_f	Steering angle	-	[rad]
α_f	Sideslip angle of front tire	-	[rad]
α_r	Sideslip angle of rear tire	-	[rad]
β	Sideslip angle of vehicle	-	[rad]
μ	Friction coefficient	-	-
m	Vehicle mass	1270	[kg]
I_z	Inertial about vehicle vertical axis	1536.7	[kg · m ²]
L_f	Distance from CG to front axle	1.05	[m]
L_r	Distance from CG to rear axle	1.895	[m]
C_f	Cornering stiffness of front tire	135000	[N/rad]
C_r	Cornering stiffness of rear tire	85000	[N/rad]

is approximated as follows:

$$F_y = \begin{cases} -C_\alpha \alpha + \frac{C_\alpha^2}{3\mu F_z} |\alpha| \alpha - \frac{1}{3} \frac{C_\alpha^3}{(3\mu F_z)^2} \alpha^3, & |\alpha| < \arctan(\frac{3\mu F_z}{C_\alpha}) \\ -\mu F_z \text{sgn} \alpha, & \text{otherwise} \end{cases} \quad (14)$$

where, the tire cornering stiffness C_α and vertical load F_z are expressed as C_f , C_r and F_{zf} , F_{zr} for the front and rear tires, respectively. In (14), the tire sideslip angles of the front and rear wheels are calculated as follows:

$$\alpha_f = \beta + \frac{L_f}{v_x} \gamma - \delta, \quad (15a)$$

$$\alpha_r = \beta - \frac{L_r}{v_x} \gamma. \quad (15b)$$

The physical meanings of the above symbols and parameters are listed in Table I. These basic intrinsic parameters m , I_z , L_f , L_r , C_f and C_r can be derived by V2V and vehicle recognition technologies presented in [18]–[21]. The friction coefficient μ can be derived from the online estimation technology [31], [32]. The longitudinal speed v_x is updated in real time by the estimation result v_p from the longitudinal estimator module.

Besides, the roads geometry parameters should also be considered as the preceding vehicle lateral motion states are related to the current vehicle position along the desired path, such as the distance error between the vehicle centre and the centreline of the road. In contrast to studies on host vehicle state estimation, the preceding vehicle steering angle is very difficult to derive. Therefore the preceding vehicle steering angle is also estimated. We assume that the preceding vehicle steering angle remains constant during the sample period, resulting in $\dot{\delta}_f = 0$.

The Serret-Frenet equations are used to describe the vehicle tracking dynamics with respect to the desired path. As shown in Fig. 2, when the vehicle drives along the desired path, there will be a lateral off set e represents the distance between vehicle center and road centerline and a heading angle ψ represents angle between tangent of road centerline and vehicle longitudinal axis. Considering the vehicle longitudinal and lateral speeds and the road curvature c , we have:

$$\dot{e} = v_x \sin \psi + v_y \cos \psi, \quad (16)$$

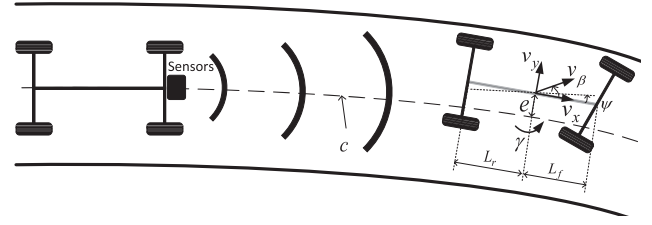


Fig. 2. Vehicle lateral motion model with respect to the desired path.

$$\dot{\psi} = \gamma - c(v_x \cos \psi - v_y \sin \psi). \quad (17)$$

It should be noted that the road curvature c is assumed to be a constant in every short sample period in this paper

$$\dot{c} = 0. \quad (18)$$

Based on the above equations, a combined preceding vehicle lateral motion model, that both the vehicle dynamic characteristics and road geometry parameters can be considered, can be derived as:

$$\dot{X} = f(X) + \bar{w}, \quad (19a)$$

$$Y = HX + \bar{\eta}, \quad (19b)$$

$$H = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \quad (19c)$$

The state vector is defined as $X = (v_y, \gamma, \delta_f, e, \psi, c)^T$, and the measurement of this system is defined as $Y = (e, \psi, c)^T$. According to [16], the lateral off set e , heading angle ψ and road curvature c can be measured with the vision system. The process noise $\bar{w}$ and measurement noise $\bar{\eta}$ represent the modelling and measurement uncertainties, respectively. The discrete state space model is consequently given as:

$$X(k+1) = T_s f[X(k)] + X(k) + \bar{w}(k), \quad (20a)$$

$$Y(k) = HX(k) + \bar{\eta}(k). \quad (20b)$$

Where the T_s is the sample time of the lateral module.

III. ESTIMATOR DESIGN

A diagram of the modular integrated estimation algorithm is shown in Fig. 3. The relative distance is detected by a radar, and the host vehicle speed is measured by a speed sensor, and the preceding vehicle intrinsic parameters are recognized with a vehicle recognition system. Moreover, the parameters e , ψ and c are measured with the vision system. The longitudinal state estimator is designed based on linear MHE and the lateral state estimator is designed based on nonlinear MHE.

A. Linear MHE for the Longitudinal Estimator

To improve the estimation accuracy and robustness to the initial guess error, a linear MHE based estimator is designed for preceding vehicle speed estimation. Eqs. (8) and (11) are adopted to describe the vehicle longitudinal motion, then linear

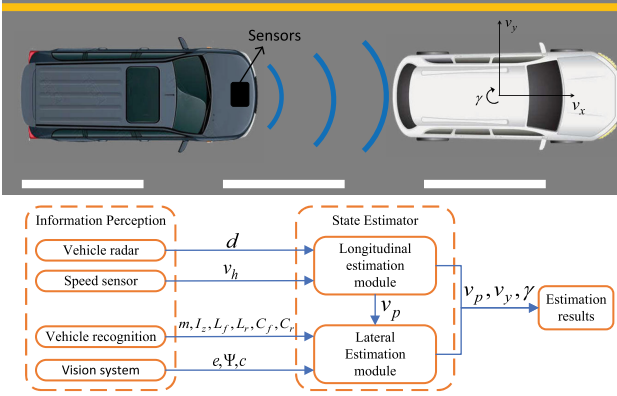


Fig. 3. Diagram of the estimators.

MHE problem for longitudinal estimation is given as follows:

$$\min_{\bar{x}, \omega, \eta} \|\bar{x}(k-n) - \bar{x}(k-n|k-n-1)\|_{P_1^{-1}(k-n|k-n-1)}^2 + \sum_{i=k-n}^{k-1} \|\omega(i)\|_{Q_1^{-1}}^2 + \sum_{i=k-n}^k \|\eta(i)\|_{R_1^{-1}}^2, \quad (21a)$$

$$s.t. \quad \bar{x}(k+1) = \bar{F}(\delta_k)\bar{x}(k) + \bar{\Gamma}(\delta_k)\omega(k), \quad (21b)$$

$$y(k) = D\bar{x}(k) + \eta(k). \quad (21c)$$

In this problem, the variables includes the state variables $\bar{x} = [\bar{x}(k-n), \dots, \bar{x}(k)]$, the process noise sequences $\omega = [\omega(k-n), \dots, \omega(k-1)]$ and the measurement noise sequences $\eta = [\eta(k-n), \dots, \eta(k)]$. The $(i|k)$ denotes the estimate at time step i with measurements until time step k . Where n is the window length of the longitudinal estimator module, and $\bar{x}(k-n|k-n-1)$ is the initial guess at time step $k-n$ which is also called as a priori state estimate. The state estimation, and process and measurement noise covariance matrixes are denoted as $P_1(k-n|k-n-1)$, Q_1 and R_1 , respectively. The $\bar{x}(k-n|k-n)$ and $P_1(k-n|k-n)$ are defined as a posteriori state estimate and error covariance at time step $k-n$, respectively. As can be seen from the above problem formulation, a large covariance Q_1 used in (21a) means a small weighting Q_1^{-1} given to the second component of (21a), which also means little trust is put on the model, and vice versa. The similar explanation can also be used in $P_1(k-n|k-n-1)$ and R_1 .

Ideally, every past measurement should be used in estimation. However, this will cause an infinite dimension optimization problem and bring unaffordable computational burden with the increasing time step. Thus, it is necessary to introduce a certain groups of measurements in the problem formulation. To this end, a horizon length n is introduced to define a window length that only $n+1$ groups of measurements in the window are used in estimation. The past measurements before time step $k-n$ should also be summarized, thus the first term in Eq. (21a) called arrival cost is introduced in problem formulation. The arrival cost is updated iteratively as the window is moving forward. Because a linear state equation is adopted to describe the vehicle longitudinal motion, the arrival cost of longitudinal estimation

is updated by the KF method [25], and the updating process are given as follows:

$$K_1(k-n) = P_1(k-n|k-n-1)D \times (R_1 + DP_1(k-n|k-n-1)D^T)^{-1}, \quad (22a)$$

$$\bar{x}(k-n|k-n) = \bar{x}(k-n|k-n-1) + K_1(k-n) \times (y(k-n) - D\bar{x}(k-n|k-n-1)), \quad (22b)$$

$$P_1(k-n|k-n) = (I - K_1(k-n)D)P_1 \times (k-n|k-n-1), \quad (22c)$$

$$\bar{x}(k-n+1|k-n) = \bar{F}(\delta_k)\bar{x}(k|k-n), \quad (22d)$$

$$P_1(k-n+1|k-n) = \bar{F}(\delta_k)P_1(k-n|k-n)\bar{F}(\delta_k)^T + \bar{\Gamma}(\delta_k)Q_1\bar{\Gamma}(\delta_k)^T. \quad (22e)$$

where K_1 is the kalman gain, I is identity matrix. It should be noted that, the above updating strategy is activated only when $t \geq n+1$. If this condition is not satisfied, there are two ways including forging faked measurements and truncating the horizon to implement MHE. The truncating horizon is adopted in this paper, that the horizon length is initialised at zero and then horizon is increased gradually to n as the time step increases. Then, the above optimization problem can be transformed into a quadratic programming problem.

B. Nonlinear MHE for the Lateral Estimator

In contrast to the vehicle longitudinal motion, a nonlinear state space equation describes the lateral motion. Combined with the nonlinear model (20), the nonlinear MHE problem for lateral estimation can be formulated as:

$$\min_{X, \bar{\omega}, \bar{\eta}} \|X(k-N) - X(k-N|k-N-1)\|_{P_2^{-1}(k-N|k-N-1)}^2 + \sum_{i=k-N}^{k-1} \|\bar{\omega}(i)\|_{Q_2^{-1}}^2 + \sum_{i=k-N}^k \|\bar{\eta}(i)\|_{R_2^{-1}}^2, \quad (23a)$$

$$s.t. \quad X(k+1) = T_s f[X(k)] + X(k) + \bar{\omega}(k), \quad (23b)$$

$$Y(k) = HX(k) + \bar{\eta}(k). \quad (23c)$$

Here the variables still includes the state variables $X = [X(k-N), \dots, X(k)]$, the process noise sequences $\bar{\omega} = [\bar{\omega}(k-N), \dots, \bar{\omega}(k-1)]$ and the measurement noise sequences $\bar{\eta}[\bar{\eta}(k-N), \dots, \bar{\eta}(k)]$. The length of the estimation window N in the lateral modular determines the number of measurement groups used in Eq. (23a). Similarly, in the lateral estimator module $X(k-N|k-N-1)$ is the initial guess at time step $k-N$ which is also called as a priori state estimate. The state estimation, and process and measurement noise covariance matrixes are denoted as $P_2(k-N|k-N-1)$, Q_2 and R_2 , respectively. The $X(k-N|k-N)$ and $P_2(k-N|k-N)$ are defined as a posteriori state estimate and error covariance at time step $k-N$, respectively. The weighting matrices

$P_2(k-N|k-N-1)$, Q_2 and R_2 have the same effect as presented in Section III-A.

In this problem, the optimization variables are the state estimation $X(k-N)$ and process noise sequences $\bar{\omega}$. Once the optimization problem is solved, the estimation at the current time step can be derived through the system state space equations. It should be noted that, the above optimization problem is not suitable for quadratic programming, but can be solved by nonlinear programming (NLP). The arrival cost can be updated by EKF and unscented KF (UKF)-based approaches, or the filtering/smoothing update strategy [28]. The EKF updating strategy is adopted in this paper, the updating process are given as follows:

$$K_2(k-N) = P_2(k-N|k-N-1)H \times (R_2 + HP_2(k-N|k-N-1)H^T)^{-1}, \quad (24a)$$

$$X(k-N|k-N) = X(k-N|k-N-1) + K_2(k-N)(Y(k-N) - HX(k-N|k-N)), \quad (24b)$$

$$P_2(k-N|k-N) = (I - K_2(k-N)H)P_2 \times (k-N|k-N-1), \quad (24c)$$

$$X(k-N+1|k-N) = T_s f[X(k-N|k-N)] + X(k-N|k-N), \quad (24d)$$

$$P_2(k-N+1|k-N) = \bar{A}(k-N|k-N)P_2(k-N|k-N) \times \bar{A}(k-N|k-N)^T + Q_2. \quad (24e)$$

where K_2 is the kalman gain, $\bar{A}(k-N|k-N)$ is the jacobian matrix which is linearised around the estimation $X(k-N|k-N)$, is defined as follow:

$$\bar{A}(k-N|k-N) = \left. \frac{\partial f}{\partial X} \right|_{X(k-N|k-N)}. \quad (25)$$

As same as the longitudinal estimator module, the truncating horizon is also adopted in the lateral estimator module.

C. Multiple Shooting for Nonlinear MHE

As discussed above, a NLP problem has to be solved in lateral estimation. When the system nonlinearity is high or the horizon length N is large, the computational time may be unacceptable for vehicle applications. This is also one of the largest obstacles preventing MHE from being applied in ADASs. Owing to the MS method, the original NLP problem can be reformulated as a sparser and less nonlinear problem compared with the single shooting (SS) method. In Fig. 4, the blue line represents real measurements, and the green dotted line represents the predicted measurements with SS-MHE, the red dotted line represents the predicted measurements with MS-MHE. In the SS-MHE method, all the predicted states and measurements propagate along the space state equation at the same given initial value $X(k-N)$. The NLP problem becomes increasingly nonlinear with increasing time step, for instance, the terminal predicted states and measurements are iteratively calculated N times,

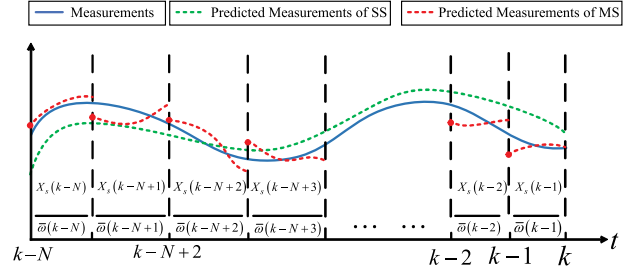


Fig. 4. Diagram of the multiple shooting method for MHE.

which makes the problem extremely nonlinear. In MS-MHE, the whole time horizon $[k-N, k]$ is divided into N segments with the sample time T_s , the j_{th} segment I_j is defined as:

$$I_j = [k-j, k-j+1], (j = 1, 2, \dots, N). \quad (26)$$

It is assumed that the dimension of process noise $\bar{\omega}$ is $n_{\bar{\omega}}$, and the i_{th} component of the process noise is $\bar{\omega}_i (i = 1, 2, \dots, n_{\bar{\omega}})$. Then the process noise can be formulated as:

$$\bar{\omega}_i(t) = \sum_{j=1}^N \bar{\omega}_{ij}(t) \chi_j(t). \quad (27)$$

Where $\bar{\omega}_{ij}(t)$ represents the process noise of the i_{th} component $\bar{\omega}_i(t)$ in segment I_j , and $\chi_j(t)$ is the unit switching function, which is defined as:

$$\chi_j(t) = \begin{cases} 1, & t \in [k-j, k-j+1] \\ 0, & t \notin [k-j, k-j+1] \end{cases}. \quad (28)$$

In this study, the process noise is parameterized by a constant in every segment:

$$\bar{\omega}_{ij}(t) \approx \sigma_{ij}. \quad (29)$$

After process parameterization of process, the states and measurements propagate in every single segment.

It should be noted that, initial values are necessary for state propagation in every segment. Hence, the initial value set is defined as $X_s = [X_s(k-N), X_s(k-N+1), \dots, X_s(k-1)]$, where the component $X_s(i) (i = k-N, k-N+1, \dots, k-1)$ denotes the initial state values in every segment. Particularly, in the first segment, the initial state value is defined as $X_s(k-N) = X(k-N)$.

To guarantee continuity of the predicted states and measurements, $N-1$ equality constraints are introduced as follows:

$$\hat{X}(i) - X_s(i) = 0, i = k-N+1, \dots, k-1. \quad (30)$$

Where $\hat{X}(K)$ is the predicted state based on the initial value $X_s(K-1)$ at time instance K .

Thus, the original NLP problem in Eq. (23) is transformed into a new one with optimization variables $\bar{\omega}$ and initial values set X_s . Then the new MHE problem can be written as:

$$\min_{X_s, \sigma, \bar{\eta}} \|X_s(k-N) - X(k-N|k-N-1)\|_{P_2^{-1}(k-N|k-N-1)}^2 + \sum_{j=1}^N \sum_{i=1}^{n_{\bar{\omega}}} \|\sigma_{ij}\|_{Q_2(i,i)^{-1}}^2 + \sum_{i=k-N}^k \|\bar{\eta}(i)\|_{R_2^{-1}}^2, \quad (31a)$$

$$s.t. \quad X(k+1) = T_s f[X(k)] + X(k) + \bar{w}(k), \quad (31b)$$

$$Y(k) = HX(k) + \bar{\eta}(k), \quad (31c)$$

$$\hat{X}(i) - X_s(i) = 0, i = k - N + 1, \dots, k - 1. \quad (31d)$$

Where $Q_2(i, i)$ ($i = 1, 2, \dots, n_{\bar{w}}$) denotes the components in the i_{th} row and column, and the optimization variables set σ is defined as:

$$\sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1N} \\ \sigma_{21} & \sigma_{22} & \dots & \sigma_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n_{\bar{w}}1} & \sigma_{n_{\bar{w}}2} & \dots & \sigma_{n_{\bar{w}}N} \end{pmatrix}. \quad (32)$$

From the above discussion, there are $n_x \times (N + 1) + n_{\bar{w}} \times N$ variables in the MS-MHE. In contrast, only $n_x + n_{\bar{w}} \times N$ variables in SS-MHE. The influence of the increase in variables will be discussed later.

D. Simulation Results

As discussed in Section III-B and III-C, NLP of lateral estimator is the main cause of heavy computational burden. To reduce the computational time, the multiple shooting method is adopted for the lateral estimator. Besides, the computational burden and horizon length N of lateral estimator are closely related. The horizon length N is also a critical parameter in MHE and directly affects the estimation accuracy. Thus, the horizon length N of lateral estimator and influence of multiple shooting will be discussed firstly in this subsection, so that some guidelines of parameter selection can be provided for the HIL experiments. To this end, a series of joint simulations with Carsim/MATLAB will be conducted and all the simulations run on a desktop computer with an Intel i7-7700 processor. It should be noted that a modified double lane change manipulation is adopted here and the longitudinal speed is set as $v_x = 30 \text{ m/s}$.

1) *Selection of the Horizon Length*: The sample time of the lateral motion model is set to $T_s = 0.007 \text{ s}$, the process and measurement noise covariance matrixes Q_2 and R_2 , the prior state guess covariance matrix $P_2(0)$, and the initial state guess $X(0)$ are set as follows:

$$Q_2 = \text{diag}(4.8E - 6, 5.1E - 17, 1.2E - 9, 4.1E - 7, 6.3E - 16, 3.7E - 13), \quad (33a)$$

$$R_2 = \text{diag}(3.2E - 3, 6.9E - 6, 4.5E - 6), \quad (33b)$$

$$P_2(0) = \text{diag}(1.5E - 5, 1E - 6, 1.7E - 7, 3.2E - 5, 1E - 7, 1.1E - 9), \quad (33c)$$

$$X(0) = (0, 0, 0, -2, 0, 0). \quad (33d)$$

It should be noted that the initial state guess is set as the true initial value. And the longitudinal speed of preceding vehicle is introduced directly from CarSim, which means that the longitudinal estimator is disabled in this simulation and only the lateral estimator is tested with different horizon length N .

To facilitate a comparison of the estimation accuracy with different values of N , the root mean square error (RMSE) and the error integration e_{int} are adopted. The error integration is

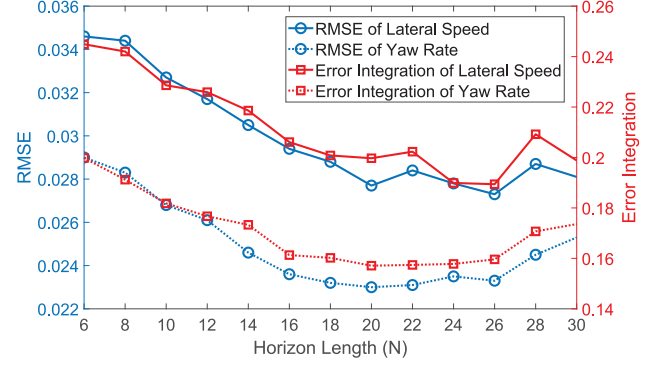


Fig. 5. RMSE and error integration for the different horizon lengths.

defined as $e_{int} = \int_{t_s}^{t_e} |err(t)| dt$, where $err(t)$ is the estimation error, and t_s and t_e are the time instants when the estimation starts and ends, respectively.

The comparison results are shown in Fig. 5, where the left and right axes represent the RMSE and error integration values, respectively. When the horizon length $N \leq 16$, both the RMSE and error integration decrease sharply with increasing horizon length. However, when the horizon length $N \geq 16$, the improvement of the estimation accuracy is not notable with increasing horizon length. This phenomenon corresponds with the conclusion derived by Rao and Rawlings in [25], that the MHE performance improves with increasing the horizon length, albeit with diminishing returns once N is sufficiently large.

However, the estimation accuracy even deteriorate when $N \geq 26$, as shown in Fig. 5, and the RMSE and error integration increase when the horizon length is excessive. This is mainly caused by the modelling error in the estimator. It can be explained that, on one hand, when N is a suitable large number, this benefits the estimation because multiple groups of measurements are used. On the other hand, as the horizon length increases, the modelling error, which can compromise the optimization process, accumulates simultaneously. As a result, diminishing returns will be obtained, and the accumulation of modelling error will even decrease the estimation accuracy when an excessively large N is adopted. Besides, the computational efficiency should also be taken into account, which means the horizon length N should not be very large. Thus, the horizon length N of lateral estimator is set as $N = 16$ in this paper, so that both the estimation accuracy and computational efficiency can be guaranteed. Similarly, the horizon length n of longitudinal estimator can be determined in the same way. Due to the limited space, the related results are not given in this paper.

2) *Influence of Multiple Shooting*: Overall, a suitable large N can improve the estimation accuracy. Meanwhile, the computational burden will also increase with the horizon length increasing. As mentioned in Section III-C, the nonlinear MHE problem are segmented into short time intervals, and the integration can be simultaneously implemented in each segment. Then, the original NLP problem is reformulated with more optimization variables as well as less nonlinear. Then the MS method is used to reduce the computational burden. The computational times with different shooting methods are shown in Fig. 6. When

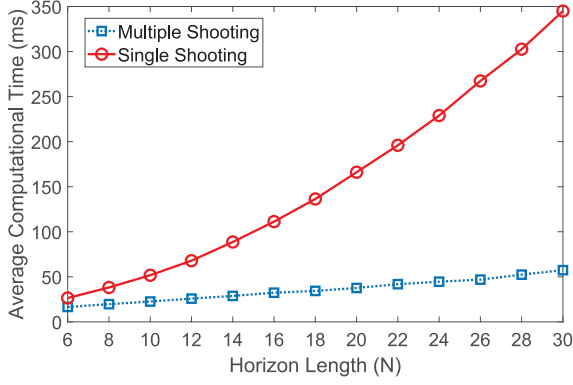


Fig. 6. Computational time with different horizon lengths.

the MS method is applied, the computational time increases linearly with the horizon length. However, the computational time increases exponentially with the horizon length, when the SS method is applied. The computational time with MS method is significantly shorter than that with the SS method.

However, more optimization variables and equality constraints are introduced in MS method to guarantee the continuity, thus making the reformulated NLP is larger as well as sparser than the original one. The increases in optimization variables and constraints are a drawback of MS method. However, the larger and sparser NLP can be solved by optimization algorithms tools such as SQP, and this makes the MS method significantly faster than the SS method [28], [29]. Thus, the proposed MHE can successfully take care of both the accuracy and computational burden.

IV. HARDWARE IN LOOP EVALUATION

As one of the key subsystems of intelligent driving, the feasibility of online estimation should be emphasized. To show the feasibility of hardware implementation and vehicular application, as well as testing the effectiveness of proposed estimators, a hardware in loop (HIL) platform, as shown in Fig. 7, was established for the experiments. Based on the simulation results given in Section III-D, a series of experiment will be conducted in this section. The HIL platform consists of two industrial personal computers (IPC), a laptop computer, a desktop computer, a Controller Area Network (CAN) bus and a dSPACE MicroAutoBox DS1401. The IPC1 and desktop computer are used to perform the vehicle dynamics simulation with CarSim as the vehicle dynamic software, IPC2 is used to provide a simulated driving environment with visual scene software SCANer. The MicroAutoBox is utilized as a rapid prototype estimator, and a laptop computer is used to monitor the whole system and store the data. Besides, CAN bus is used to transmit data among the above devices.

A. Longitudinal Estimation Results

In this experiment, the preceding vehicle starts 100 m from the starting point and drives at an initial speed of 20 m/s. The host vehicle accelerates to 30 m/s at the starting point, and then approaches and follows the preceding vehicle. To test the

TABLE II
LONGITUDINAL ESTIMATION RESULTS COMPARISON

States and Estimators		RMSE	e_{max}	t_{cog}
v_p	MHE	0.1954	0.9943	4.8
	KF	0.4552	1.6927	8.5
C_4	MHE	0.0627	0.3031	4.6
	KF	0.1747	0.753	19

effectiveness of the longitudinal speed estimator, two scenarios are established in the experiments. In scenario 1, the preceding vehicle will accelerate and decelerate and maintain a constant speed between acceleration and deceleration, while a time-varying speed profile is established between acceleration and deceleration in the scenario 2. The speed and relative distance of the two vehicles are shown in Fig. 8.

In this experiment, the sample time $\delta_k = 0.05$ s, and the measurements are polluted by artificial noise following a normal distribution with zero mean, and standard deviations 0.5 m and 0.01 m/s for the relative distance and host vehicle speed, respectively. According to [15], the coefficients in Eq. (1) are set to $C_1 = 0.5$, $C_2 = 0.125$ and $C_3 = -0.125$. The parameter C_4 is an indicator of the relative distance between the host and preceding vehicle when the vehicles enter a steady-state equilibrium. For example, under the assumption $C_4 = -2.5$, the vehicles drive in a steady-state equilibrium at $v_x = 20$ m/s, the relative distance is $d = 40$ m, for $v_x = 30$ m/s, the relative distance is $d = 50$ m. It should be noted that, the aggressiveness parameter can not be known in advance, so a mean value $C_4 = -2.5$ [15] is considered as an in the initial state guess. The estimator works when the relative distance is smaller than 100 m in the simulations. The horizon length of longitudinal estimator module is set as $n = 10$, which is determined by the way mentioned in Section III-D. To compare the performance of MHE and KF, the model parameters remains the same. The process and measurement noise covariance matrixes Q_1 and R_1 , the prior state guess covariance matrix $P_1(0)$, and the initial state guess $\bar{x}(0)$ are set as:

$$Q_1 = \text{diag}(0.5, 0.5)^2, \quad (34a)$$

$$R_1 = \text{diag}(0.5, 0.01)^2, \quad (34b)$$

$$P_1(0) = \text{diag}(5, 0.5, 0.1, 0.001, 1.25)^2, \quad (34c)$$

$$\bar{x}(0) = (30 \ 100 \ 30 \ 0 \ -2.5)^T. \quad (34d)$$

The estimation results are shown in Fig. 9 and summarized in Table II. When the relative distance is larger than 100 m, the estimator does not work and the output is set to 30 m/s. At approximately 12 s, the estimator starts to estimate the preceding vehicle speed according to the measurements. To compare the performance of KF and MHE, the estimation results are considered to converge to the true values if they satisfy the following:

$$|v_p(k|k) - v_p(k)| \leq 0.2, \quad (35a)$$

$$|C_4(k|k) - C_4(k)| \leq 0.2. \quad (35b)$$

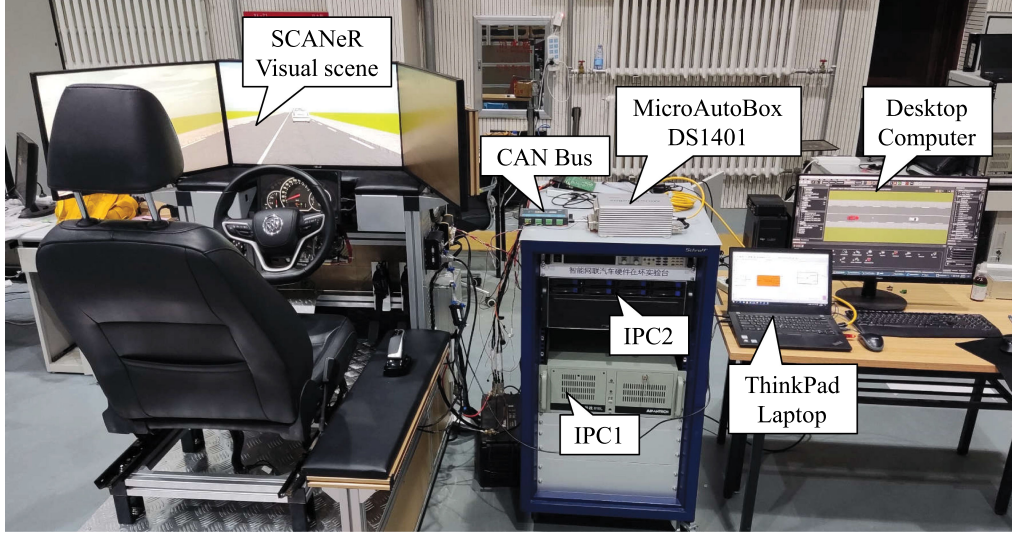


Fig. 7. Hardware in loop experiment platform.

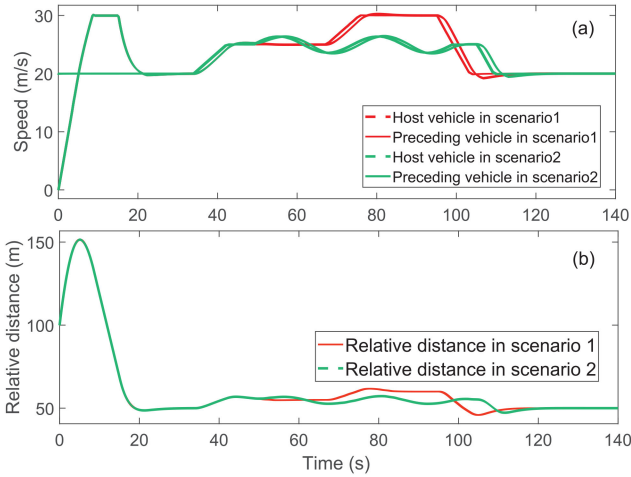


Fig. 8. The speed and relative distance between the vehicles: (a) speed profiles of the two vehicles, (b) relative distance between the two vehicles.

A time index t_{cog} is defined to represent the time period to the convergence time. It should be noted that, an initial guess should be given in advance. However, an exactly accurate initial guess is very difficult to be obtained, it is very necessary to test the converge speed under initial guess error. The speed estimation results with large initial error are shown in Fig. 9(a), after a period of time, both the KF and MHE converge to the true value. However, the KF requires more time to converge to the true value than the MHE. This can be explained by the fact that, the KF utilizes only one current measurement to estimate the speed. In contrast, $n + 1$ measurements are utilized in the MHE for the speed estimation, which means that, a poor initial guess plays a less important role in MHE. For comparison, the speed estimation results with accurate initial guess are given in Fig. 9(b), both the KF and MHE can achieve a high estimation accuracy. In addition, both the root mean square error (RMSE) and maximum error e_{max} of MHE are smaller than those of the KF. Similar results are shown in Figs. 9(c) and 12.

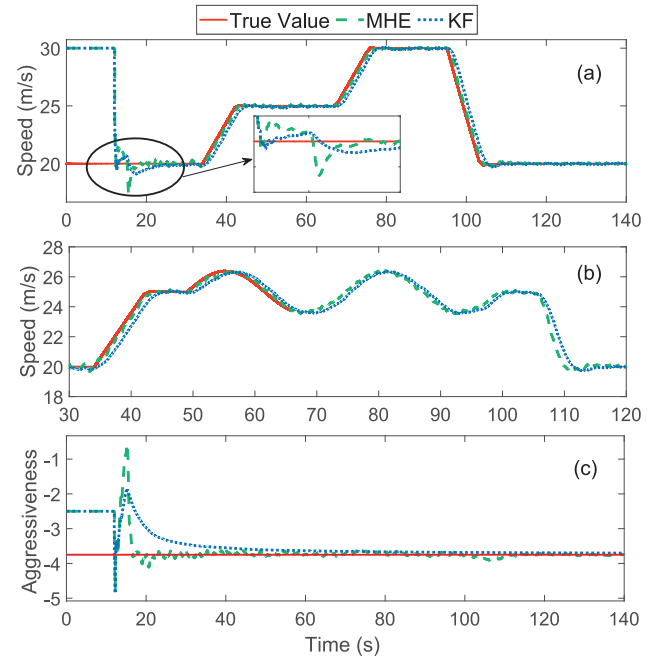


Fig. 9. Estimation results of the longitudinal module: (a) estimation results for scenario 1; (b) estimation results for scenario 2; (c) estimation results of the driver aggressiveness.

B. Lateral Estimation Results

Based on the estimation result of preceding vehicle speed, the longitudinal speed adopted in lateral estimator module can be updated. To fully test the proposed MHE-based lateral estimator performance, the preceding vehicle will perform double lane change manipulations under different conditions, which including low-speed, accelerating, high-speed and decelerating conditions. The speed profile of the preceding vehicle is shown in Fig. 9(a).

To highlight the superiority of the MHE-based estimator proposed in this paper, EKF-based and UKF-based estimators

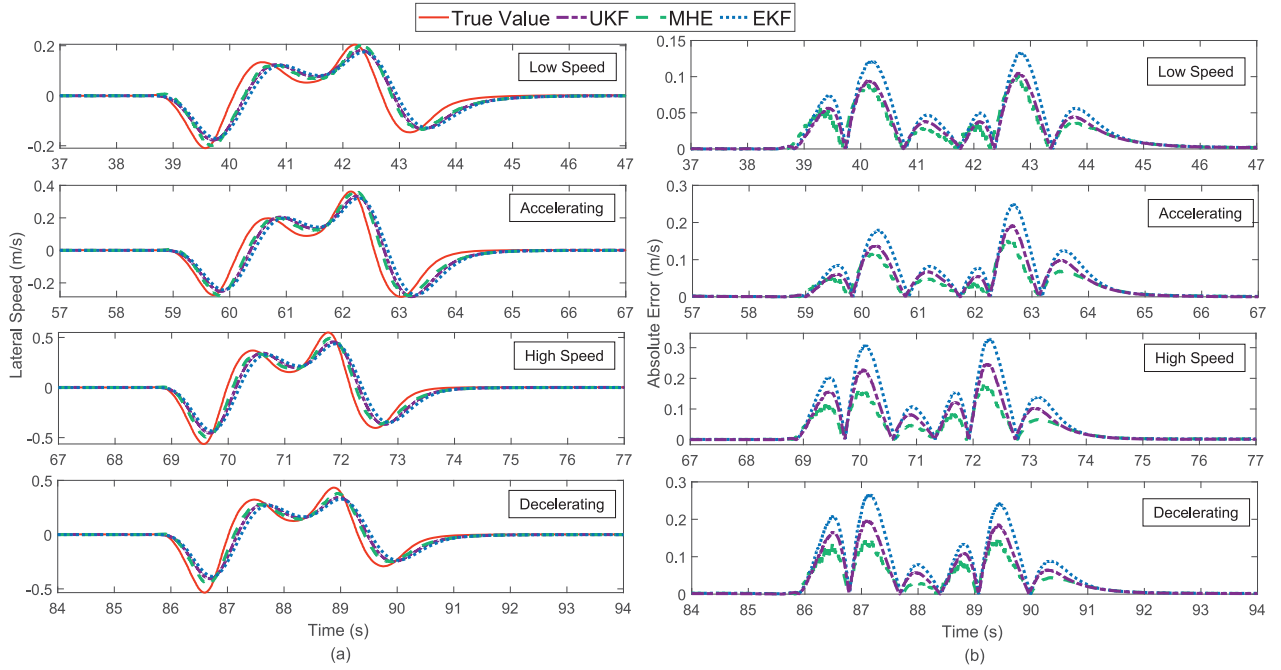


Fig. 10. Estimation results of lateral speed under DLC scenario: (a) estimation results of lateral speed, (b) absolute error of estimated lateral speed.

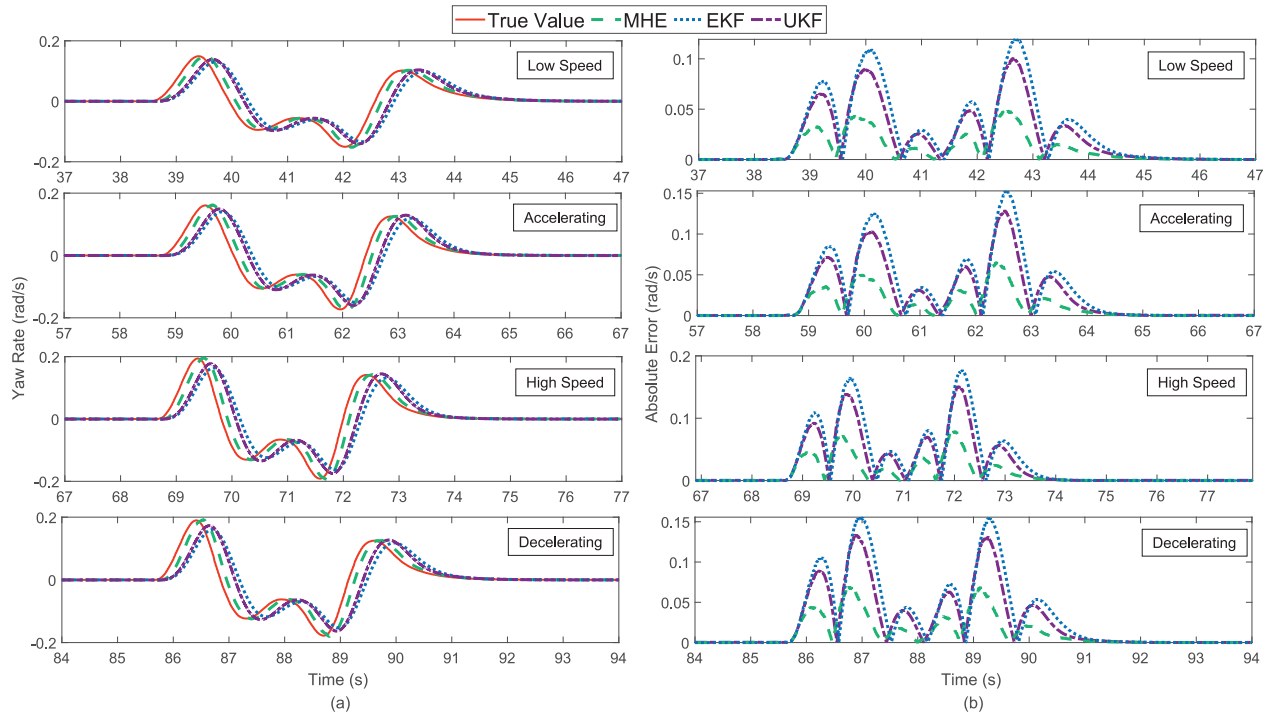


Fig. 11. Estimation results of yaw rate under DLC scenario: (a) estimation results of yaw rate, (b) absolute error of estimated yaw rate.

are designed with the same parameters for the fair comparison. In the UKF-based estimator, a simplified strategy is adopted. The number of sigma point is set to be 12, and all the weighting factors of sigma points are set as $w_i = 1/12$, where $i = 1, 2, \dots, 12$. Then, a standard UKF update process is used to derive the estimation results. Only the estimation accuracy is examined here, thus the parameters and initial conditions are consistent

with Section III-D. Convergence with a large initial error and robustness to measurement disturbance will be discussed later.

The estimation results are shown in Figs. 10 and 11. The lateral speed and its estimation are shown in Fig. 10(a). Both the EKF and MHE results track the true value well when the longitudinal speed is varied under the four different conditions. There is a short estimation delay between the estimation and true values,

TABLE III
ESTIMATION RESULTS COMPARISON

States and Estimator		Low Speed		Accelerating		High Speed		Decelerating		Initial Error	Disturbance
		RMSE	e_{max}	RMSE	e_{max}	RMSE	e_{max}	RMSE	e_{max}	t_{cog}	t_{cog}
v_y	MHE	0.0333	0.1019	0.0453	0.1548	0.0556	0.1860	0.0505	0.1449	0.64	1.1
	EKF	0.0465	0.1325	0.0750	0.2515	0.1080	0.3305	0.0907	0.2678	1.89	1.8
	P_{EKF}	28.4%	23.1%	39.6%	38.4%	48.5%	43.7%	44.3%	45.8%	66.1%	38.8%
	UKF	0.0359	0.1049	0.0568	0.1934	0.0794	0.2486	0.0678	0.1980	1.75	25
	P_{UKF}	7.2%	2.9%	20.2%	19.9%	30.0%	25.2%	25.5%	26.8%	63.4%	95.6%
γ	MHE	0.0169	0.0491	0.0196	0.0667	0.0236	0.0796	0.0219	0.0696	0.81	0.9
	EKF	0.0427	0.1194	0.0489	0.1529	0.0575	0.1772	0.0538	0.1557	1.63	1.4
	P_{EKF}	60.4%	58.8%	59.9%	56.3%	58.9%	55.1%	59.3%	55.3%	50.3%	35.7%
	UKF	0.0351	0.1003	0.0404	0.1292	0.0479	0.1515	0.0447	0.1344	1.42	25.5
	P_{UKF}	51.8%	51.0%	51.5%	48.4%	50.7%	47.5%	51.0%	48.2%	42.9%	96.5%

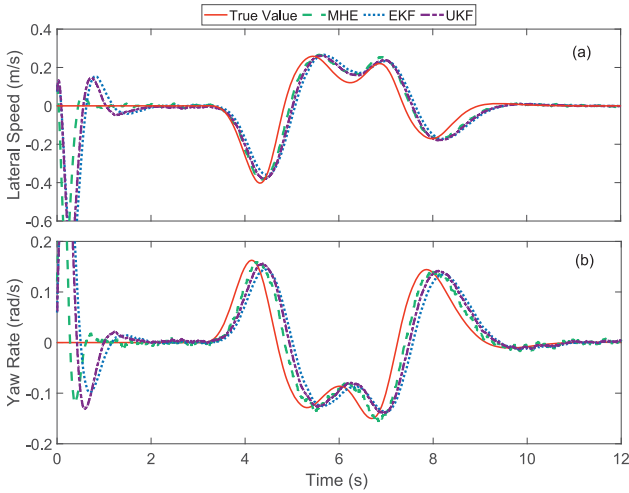


Fig. 12. Estimation results for a large initial guess error: (a) estimation results of lateral speed, (b) estimation results of yaw rate.

which are approximately 0.1 s for MHE, 0.16 s for UKF, and 0.21 s for EKF. A shorter estimation delay implies a higher accuracy. Meanwhile the absolute error of lateral speed estimation is shown in Fig. 10(b), and the MHE absolute error is below that of UKF and EKF. Similar phenomena are also observed in the yaw rate estimation results, as shown in Fig. 11(a). As can be seen in estimation results, estimators can track the real values well under the four different conditions, and the estimation delays are approximately 0.11 s for MHE, 0.25 s for UKF, and 0.32 s for EKF. The yaw rate estimation absolute error, which is given in Fig. 11(b), also reveals that MHE has a better accuracy than UKF and EKF.

To further investigate and compare the performance of MHE, UKF and EKF, two performance improvement indexes of EKF and UKF are defined as:

$$P_{EKF} = \frac{Index_{EKF} - Index_{MHE}}{Index_{EKF}} \times 100\%, \quad (36a)$$

$$P_{UKF} = \frac{Index_{UKF} - Index_{MHE}}{Index_{UKF}} \times 100\%. \quad (36b)$$

Where $Index_{EKF}$, $Index_{UKF}$ and $Index_{MHE}$ are statistical indexes, such as the RMSE, maximum absolute error e_{max} and convergence time t_{cog} , of the EKF, UKF and MHE, respectively.

The statistical results of the estimation are summarized in Table III.

From the aforementioned discussion, it can be concluded that the MHE-based estimator has a much higher accuracy than the UKF-based and EKF-based estimators. Besides, due to the nonlinear model is adopted, the UKF-based estimator also has a better performance in accuracy than EKF-based estimator. This can be explained by the fact that, on the one hand, the EKF relies on a linearized approximate model near the operation point, and in contrast, a nonlinear model, which describes the system better, is used in MHE and UKF. On the other hand, there are multiple groups of measurements used for estimation in MHE, in contrast, only one group of measurement is used in the EKF and UKF, which means MHE has a deeper degree of optimization and makes MHE more robust to modelling uncertainties and inaccuracies.

C. Robustness Test

In this subsection, the robustness of estimators will be tested. Three common conditions including large initial guess error, measurement disturbances and parameter uncertainty are considered.

1) *Large Initial Guess Error*: In real applications, an initial state guess should be provided first before the estimation, and it is impossible to attain a completely accurate guess. It is very common that there is a large error between the given initial guess and the true value. Instead of the accurate guess used in Section IV, an initial guess with a large error is defined as $X(0) = (0.09, 0.06, 0.1, 0, 0, 0)^T$. The other parameters are set as same as in Section IV, and the horizon length and sample time are set as $N = 16$ and $T_s = 0.007$ s, respectively. And the convergence time t_{cog} is defined as the time from giving the initial guess to obtaining an accurate value.

The comparison results are shown in Fig. 12 and listed in Table III. As shown in Fig. 12, for a large initial guess error, the estimation results will first abruptly increase in the opposite direction. Then, the estimation results decrease and converge to the true value after a short period of time. The convergence speed of MHE is much higher than that of UKF and EKF. According to the statistical results summarized in Table III, for the lateral speed estimation, it takes approximately 0.64 s, 1.75 s and 1.1 s to converge for MHE, UKF and EKF, respectively.

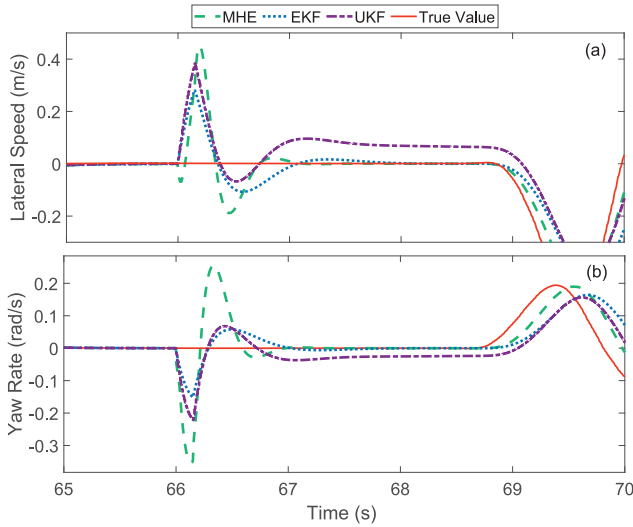


Fig. 13. Estimation results with a measurement disturbance: (a) estimation results of lateral speed, (b) estimation results of yaw rate.

For the yaw rate estimation, approximately 0.81 s, 1.42 s and 1.63 s are required by the MHE, UKF and EKF, respectively, to converge.

At the beginning, the measurements used in MHE, UKF and EKF are similar, then the result is that their estimation accuracies are very similar. The phenomenon also exists when the horizon length of MHE is very small. As more measurements are utilized in MHE, the inaccurate initial guess doesn't play an important role in the estimation. Then the estimated values in MHE can converge to the true value much quickly.

2) *Measurement Disturbances*: It is also very common that measurements may be polluted by unexpected disturbances. Therefore, a good estimator should have a high robustness to the measurement disturbances. To test the robustness of estimators to external measurement disturbances, an external measurement disturbance $y_{dis} = (3, 0.3, 5E - 2)^T$ will be injected into the system at time instant $t = 66$ s. The other parameters are kept the same as in Section IV. The convergence time t_{cog} here is defined as the time period from measurement disturbance injection to estimation converge to the true value.

The comparison results are shown in Fig. 13 and listed in Table III. After the disturbance is injected into the system, both the MHE, UKF and EKF estimation results deviate from the true values. After a small period of time, both the MHE and EKF can converge to the true value, and the MHE convergence speed is also much higher than that of the EKF. As indicated in Table III, the EKF requires 1.8 s and 1.4 s for the lateral speed and yaw rate estimation results converging to the true values, while in contrast, MHE only requires 1.1 s and 0.9 s, respectively. Compared with the MHE and EKF, UKF has a very weak convergence ability for the external measurement disturbance, it takes more than 25 s to converge the true value when the measurement is polluted by disturbance.

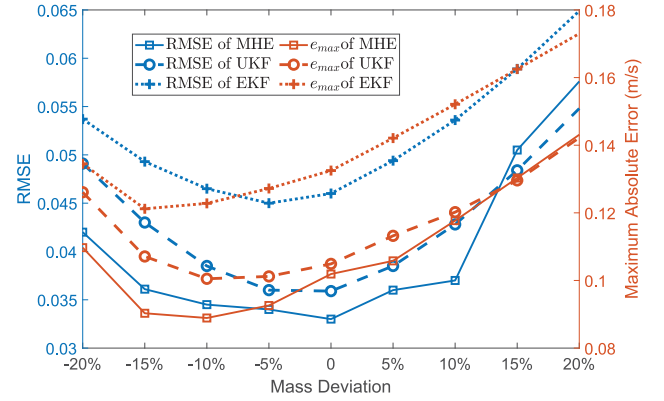


Fig. 14. RMSE and e_{max} of lateral speed for the different mass deviation.

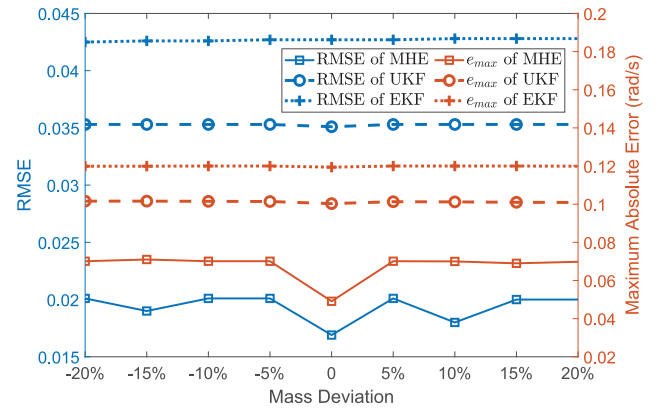


Fig. 15. RMSE and e_{max} of yaw rate for the different mass deviation.

As discussed above, only one group of measurements is used in UKF and EKF, and when the disturbance is introduced, only a polluted measurement can be derived by them. However based on the objective function Eq. (23), the estimation results in MHE is partly influenced by the polluted measurement. In other words, the current polluted measurement is less disruptive in MHE compared to UKF and EKF.

3) *Parameter Uncertainty*: In this paper, the model parameters used in lateral estimator module are derived vehicle recognition technologies and V2V. Thus, only the basic intrinsic parameters can be obtained. However, some of these parameters may vary under different scenarios. For example, the mass of vehicle varies with different number of passengers. And a group of experiment is constructed to test the robustness of MHE estimator to the parameter uncertainty. In this paper, due to the limited space, only the mass deviation is discussed here.

The range of mass deviation is set from -20% to 20% of the basic mass. Two indices including RMSE and maximum absolute error e_{max} of MHE and EKF with different mass deviations are compared. The results of lateral speed are given in Fig. 14. Both the RMSE and e_{max} increase with the increasing mass deviation. And a very small increase can be observed when the mass deviation is kept within $\pm 10\%$. The results show that the presented estimators has a good robustness to parameter uncertainties. The results of yaw rate are given in

Fig. 15, it can be seen that the estimators performance of yaw rate estimation is almost immune to mass deviation. This can be explained that the direct measurement of the preceding vehicle's heading angle makes it possible to estimate the yaw rate directly without relying on the vehicle dynamics. It should be noted that, the presented MHE estimator still has a better performance and robustness than the UKF and EKF based estimators with different mass deviations.

V. CONCLUSION

In this paper, a modular integrated estimation scheme has been proposed for the preceding vehicle longitudinal and lateral states. With the consideration of different driver aggressiveness and model uncertainties, a linear moving horizon estimator was designed to estimate the vehicle longitudinal speed. In order to improve estimation accuracy, a nonlinear vehicle model was adopted, then a nonlinear moving horizon estimator with MS method was designed to estimate the vehicle lateral states. Compared with SS method, the proposed MHE algorithm has a shorter computational time by revising the objective function. A series of simulations and HIL experiments were conducted to test feasibility of hardware implementation and vehicular application, as well as the effectiveness of the proposed estimators, the experiment results showed that the proposed estimators can provide an accurate estimate of preceding vehicle states. For a large initial guess error and external measurement disturbance, MHE have a much higher convergence speed and robustness than UKF and EKF. Due to the limited experimental conditions, no real vehicle data is tested in this study. Our future research will be concentrated on the more extensive experimental validation of the proposed MHE estimator.

REFERENCES

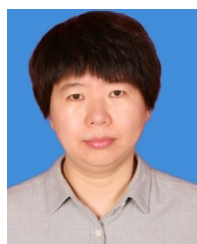
- [1] S. Patole, M. Torlak, D. Wang, and M. Ali, "Automotive radars: A review of signal processing techniques," *IEEE Signal Process. Mag.*, vol. 34, no. 2, pp. 22–35, Mar. 2017.
- [2] S. E. Shladover, "Connected and automated vehicle systems: Introduction and overview," *J. Intell. Transp. Syst.*, vol. 22, no. 3, pp. 190–200, 2018.
- [3] L. Zhang *et al.*, "An adaptive backstepping sliding mode controller to improve vehicle maneuverability and stability via torque vectoring control," *IEEE Trans. Veh. Technol.*, vol. 69, no. 3, pp. 2598–2612, Mar. 2020.
- [4] H. Guo, D. Cao, H. Chen, C. Lv, H. Wang, and S. Yang, "Vehicle dynamic state estimation: State of the art schemes and perspectives," *IEEE/CAA J. Automat. Sinica.*, vol. 5, no. 2, pp. 418–431, Mar. 2018.
- [5] X. Jin *et al.*, "Online estimation of inertial parameter for lightweight electric vehicle using dual unscented kalman filter approach," *IET Intell. Transp. Syst.*, vol. 14, no. 5, pp. 412–422, 2020.
- [6] A. Abosekeen *et al.*, "Adaptive Cruise Control Radar-Based Positioning in GNSS Challenging Environment," *IET Radar Sonar and Navigation.*, vol. 13, no. 10, pp. 1666–1677, 2019.
- [7] L. Guoqiang, and D. Gorges, "Ecological adaptive cruise control and energy management strategy for hybrid electric vehicles based on heuristic dynamic programming," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 9, pp. 3526–3535, Sep. 2019.
- [8] W. Yang *et al.*, "Research on longitudinal active collision avoidance of autonomous emergency braking pedestrian system (AEB-P)," *SENSORS.*, vol. 19, no. 21, 2019, Art. no. 4671.
- [9] T. Kirubarajan, Y. Bar-Shalom, K. R. Pattipati, and I. Kadar, "Ground target tracking with variable structure IMM estimator," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 36, no. 1, pp. 26–46, Jan. 2000.
- [10] M. Ulmke, and W. Koch, "Road-map assisted ground moving target tracking," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 42, no. 4, pp. 1264–1274, Oct. 2006.
- [11] Z. Liu *et al.*, "Vision-based inter-vehicle distance estimation for driver alarm system," *IET Intell. Transp. Syst.*, vol. 13, no. 6, pp. 927–932, 2019.
- [12] D. Helbing, P. Molnar, "Social force model for pedestrian dynamics," *Phys. Rev. E., Statist. Phys. Plasmas Fluids Relat. Interdiscip. Top.*, vol. 51, pp. 4282–4286, 1995.
- [13] R. Ding, M. Yu, H. Oh, and W. Chen, "New multiple-target tracking strategy using domain knowledge and optimization," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 47, no. 4, pp. 605–616, Apr. 2017.
- [14] R. M. Jannes *et al.*, "Characterizing the impact of production adaptive cruise control on traffic flow: An investigation," *Tras. B-Transp. Dynas.*, vol. 7, no. 1, pp. 992–1012, 2019.
- [15] D. Song, R. Tharmarasa, T. Kirubarajan, and X. N. Fernando, "Multi-vehicle tracking with road maps and car-following models," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 5, pp. 1375–1386, May 2018.
- [16] Y. Wang, Z. Zhou, C. Wei, Y. Liu, and C. Yin, "Host-target vehicle model-based lateral state estimation for preceding target vehicles considering measurement delay," *IEEE Trans. Ind. Inform.*, vol. 14, no. 9, pp. 4190–4199, Sep. 2018.
- [17] J. Eilbrecht and O. Stursberg, "Hierarchical solution of non-convex optimal control problems with application to autonomous driving," *Eur. J. Control.*, vol. 50, pp. 188–197, 2019.
- [18] S. Wei, Y. Zou, X. Zhang, T. Zhang, and X. Li, "An integrated longitudinal and lateral vehicle following control system with radar and vehicle-to-vehicle communication," *IEEE Trans. Veh. Technol.*, vol. 68, no. 2, pp. 1116–1127, Feb. 2019.
- [19] B. Zhang, "Reliable classification of vehicle types based on cascade classifier ensembles," *IEEE Trans. Intell. Transp. Syst.*, vol. 14, no. 1, pp. 322–332, Mar. 2013.
- [20] R. Baran, A. Glowacz, and A. Matiolanski, "The efficient real- and nonreal-time make and model recognition of cars," *Multimedia Tools Appl.*, vol. 74, no. 12, pp. 4269–4288, 2015.
- [21] J. Fang, Y. Zhou, Y. Yu, and S. Du, "Fine-grained vehicle model recognition using a coarse-to-fine convolutional neural network architecture," *IEEE Trans. Intell. Transp. Syst.*, vol. 18, no. 7, pp. 1782–1792, Jul. 2017.
- [22] H. Khayyam, "Stochastic models of road geometry and wind condition for vehicle energy management and control," *IEEE Trans. Veh. Technol.*, vol. 62, no. 1, pp. 61–68, Jan. 2013.
- [23] S. Milani, H. Khayyam, H. Marzbani, W. Melek, N. L. Azad and R. N. Jazar, "Smart autodriver algorithm for real-time autonomous vehicle trajectory control," *IEEE Trans. Intell. Transp. Syst.*, to be published, doi: 10.1109/TITS.2020.3030236.
- [24] W. Ping, L. Hanghang, Z. Lin, D. Haitao and C. Hong, "Design and experimental verification of real-time nonlinear predictive controller for improving the stability of production vehicles," *IEEE Trans. Control. Syst. Technol.*, vol. 29, no. 5, pp. 2206–2213, Sep. 2021.
- [25] C. V. Rao and J. B. Rawlings "Constrained process monitoring: Moving horizon approach," *AIChE J.*, 2002, vol. 48, no. 1, pp. 97–109
- [26] M. Canale, L. Fagiano, and C. Novara, "A direct moving horizon approach to vehicle side-slip angle estimation," in *Proc. 49th IEEE Conf. Decis. Control*, 2010, pp. 2898–2903
- [27] M. Zanon, J. Frasch, and M. Diehl, "Nonlinear moving horizon estimation for combined state and friction coefficient estimation in autonomous driving," in *Proc. Eur. Control Conf.*, 2013, pp. 4130–4135
- [28] W. Zhang, Z. Wang, C. Zou, L. Drugge and M. Nybacka, "Advanced vehicle state monitoring: Evaluating moving horizon estimators and unscented kalman filter," *IEEE Trans. Veh. Technol.*, vol. 68, no. 6, pp. 5430–5442, Jun. 2019.
- [29] L. T. Biegler, *Nonlinear Programming: Concepts, Algorithms, and Applications to Chemical Processes*. Philadelphia, PA, USA: SIAM, 2010
- [30] H. B. Pacejka, *Tyre and Vehicle Dynamics*. Oxford, U.K.: Butterworth-Heinemann, 2012.
- [31] C. Ahn, and H. E. Tseng, "Robust estimation of road friction coefficient using lateral and longitudinal vehicle dynamics," *Veh. System Dyn.* vol. 50, no. 6, pp. 1–25, 2012.
- [32] X. Xia *et al.*, "Estimation of maximum road friction coefficient based on lyapunov method," *Int. J. Automot. Technol.* vol. 17, no. 6, pp. 991–1002, 2016.
- [33] W. Helly, "Simulation of bottlenecks in single lane traffic flow," in *Proc. Theory Traffic Flow*, 1959, pp. 207–238.
- [34] L. Zhang *et al.*, "Human-centered torque vectoring control for distributed drive electric considering driving characteristics," *IEEE Trans. Veh. Technol.*, vol. 70, no. 8, pp. 7386–7399, Aug. 2021.



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